

Status of Audenaert–Kittaneh Problem 4

An explicit 2014 disproof and a supplementary exact 2×2 witness

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Status: literature_already_answered.

Source question

In Problem 4 on page 3 of Audenaert and Kittaneh, arXiv:1201.5232v3 [1], the authors ask whether, for all positive semidefinite matrices A, B , every $p > 0$, and every singular-value index i ,

$$\sigma_i(A^p B^{1-p} + B^p A^{1-p}) \leq \sigma_i(A + B). \quad (1)$$

Here singular values are arranged in non-increasing order.

Exact literature answer

Bottazzi, Elencwajg, Larotonda, and Varela explicitly identify this question in Section 3 of arXiv:1403.7472 [2]. Their Example 3.2 gives positive definite 3×3 matrices A, B at $p = 1/2$ satisfying

$$\sigma_3(A^{1/2} B^{1/2} + B^{1/2} A^{1/2}) = 591.716 > 443.017 = \sigma_3(A + B).$$

Thus (1) is false. The supporting authors explicitly cite Audenaert–Kittaneh Problem 4 and state that the example answers it negatively, so this is an exact, provenance-aware literature resolution rather than an inferred implication.

Supplementary exact 2×2 certificate

The direct attack in this run independently found a smaller symbolic witness. Let

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}, \quad A = 64P, \quad B = Q, \quad p = \frac{1}{3}.$$

Both P and Q are orthogonal rank-one projections, hence $A, B \geq 0$, $A^{1/3} = 4P$, $A^{2/3} = 16P$, and $B^{1/3} = B^{2/3} = Q$. Therefore

$$L := A^{1/3} B^{2/3} + B^{1/3} A^{2/3} = 4PQ + 16QP = \begin{pmatrix} 5 & \sqrt{3} \\ 4\sqrt{3} & 0 \end{pmatrix}.$$

The eigenvalues of L^*L are $38 \pm 10\sqrt{13}$. Consequently

$$\sigma_2(L) = \sqrt{38 - 10\sqrt{13}} > 1,$$

because $37 > 10\sqrt{13}$. Meanwhile

$$A + B = \begin{pmatrix} 257/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}, \quad \sigma_2(A + B) = \frac{65 - \sqrt{4033}}{2} < 1,$$

where the last inequality follows from $4033 > 63^2$. Hence already in dimension two,

$$\sigma_2\left(A^{1/3}B^{2/3} + B^{1/3}A^{2/3}\right) > \sigma_2(A + B).$$

This exact certificate is included as a transparent strengthening of the recorded negative answer; it is not presented as the first resolution of Problem 4.

Search evidence and scope

The run searched its lightweight registry, solution, attempt, and proof-gap indexes by arXiv id and core formula keywords, then searched arXiv/general scholarly results by the exact product formula, the source title/authors, and “singular value” plus “counterexample.” This found arXiv:1403.7472, whose abstract and Example 3.2 directly identify the source problem. No claim is made about the novelty of the supplementary 2×2 simplification. The conclusion here concerns only Problem 4; the many other questions collected in arXiv:1201.5232 are outside this packet.

References

- [1] K. M. R. Audenaert and F. Kittaneh, *Problems and Conjectures in Matrix and Operator Inequalities*, arXiv:1201.5232v3 (2012).
- [2] T. Bottazzi, R. Elencwajg, G. Larotonda, and A. Varela, *Inequalities related to Bourin and Heinz means with a complex parameter*, arXiv:1403.7472 (2014), Section 3, Example 3.2.